Smartphone-Integrated AI Microscopy for Tuberculosis and Scalable Disease Diagnostics

RESEARCH PROJECT PROPOSAL & 12-MONTH IMPLEMENTATION FRAMEWORK

1. Executive Summary

Tuberculosis (TB) remains a major global health challenge, with diagnosis in resource-constrained regions continuing to rely on manual sputum smear microscopy, which is labor-intensive, time-consuming, and susceptible to human error. To address these limitations, this project will develop and evaluate a cost-effective, smartphone-integrated machine learning application designed to capture microscopic fields through an optical viewing lens and automatically detect ***Mycobacterium tuberculosis*** bacilli in real time. While the immediate priority will focus on tuberculosis detection as a foundational use case, the architectural framework of the mobile application will be designed for scalability to other microscope-viewable conditions such as malaria and HIV-related opportunistic infections.

2. Abstract & Technical Framework

A mobile application will be developed to interface with a smartphone camera mounted onto a standard light microscope via a mechanical adapter, allowing sputum smear slides prepared using acid-fast staining to be imaged directly through the eyepiece. A lightweight computer vision and machine learning model, such as YOLOv8 or a custom Convolutional Neural Network, will be trained on a dataset of mobile-captured microscopic images to identify, segment, and classify rod-shaped bacillus structures, with performance evaluated using sensitivity, specificity, and mean average precision.

The mobile application will successfully process frames captured via the smartphone lens, achieving high object-level detection precision, strong recall, and reliable overall accuracy, while demonstrating the capability to flag positive fields of view within seconds to significantly reduce preliminary screening time per slide. Ultimately, integrating machine learning into a smartphone-based microscopy framework will provide a viable, portable, and rapid screening tool that bridges diagnostic gaps in peripheral healthcare facilities and expedites prompt clinical intervention.

3. Scalability & Future Scope

Beyond tuberculosis, the modular architecture of the diagnostic software will support future expansion into:

- **Malaria Diagnostics:** Automated detection of plasmodium parasites in Giemsa-stained blood smears.
- **HIV-Related Conditions:** Screening for opportunistic infections observable via standard microscopic analysis.

4. Implementation & Research Objectives (12-Month Timeline)

The project is structured across a 12-month period with an estimated budget allocation of $15,000:

Phase & Timeline	Core Objectives	Expected Outcomes
Phase 1: Months 1–3 Budget: $3,500	Design 3D-printed optical mechanical adapters for microscope-smartphone alignment; collect and curate initial dataset of mobile-imaged sputum smears.	Prototyped adapters; standardized dataset of mobile-captured microscopic images.
Phase 2: Months 4–7 Budget: $4,000	Train and optimize lightweight computer vision models (YOLOv8 / CNN); develop core mobile application interface for live camera feed processing.	Trained ML model meeting target accuracy metrics; functional mobile application prototype.
Phase 3: Months 8–10 Budget: $4,500	Conduct laboratory performance testing, comparative benchmarking against manual microscopy, and software refinement.	Validation report quantifying precision, recall, and processing latency; iterative bug fixes.
Phase 4: Months 11–12 Budget: $3,000	Execute pilot testing in clinical/peripheral settings; compile documentation, technical manuals, and research manuscripts for dissemination.	Field evaluation report; completed technical report and publication drafts.

5. Institutional Collaboration

Pilot implementation and testing are proposed to be conducted in collaboration with reference laboratory facilities, such as the National Tuberculosis Reference Laboratory (NTRL), to ensure clinical relevance and rigorous validation.